Element A

Many different animals rely on humans to keep them alive and take care of their well-being. One of these animal species people commonly keep as pets across the nation are different types of reptiles in the united states it is reported through veterinary records over 13.4 million reptiles are kept as pets in the united states(https://secrettohappypets.com). Some of these caretakers are well experienced and educated on what it takes to make a healthy and sustainable habit for a reptile to live in, others are not educated on reptiles are kept in a habit that does not support there cold blooded needs.

One of the most commonly owned reptiles across the world are many different breeds of turtles with over 5 million turtles owned as pets in the unites states alone. Many household have very limited knowledge on turtles and reptiles before they go out and buy them as pets because there seen as low maintenance and easy to take care of. Turtles are very different animals, there are many different breeds who each need their own unique habitat some turtles only live in the water and other living in the Sahara desert and rarely ever go into the water. There cold blooded animals which mean they rely on the outside temperature to heat there body and they are not able to self-regulate their body temperature. These different aspects of turtles presents a different and unique challenge in housing them as pets.

At Towson University we are home to two different breeds of turtles, a female red eyed slider and a male western painted shell. After talking with one of our primary stake holders Robert we learned that a problem he noticed was that the female turtle who is much larger tends to push the smaller male turtle out of the basking area. There tank is 25 inch tall 48.5 inches in length and a width of 12.5 inches. There Basking area is 15.5in x 13in x 10in. This basking area is technically big enough to hold both of the turtles are the same time, but turtles will compete for the best basking area and in turtles the bigger one always wins.

Even though the basking area is largest enough for the turtles there is worry that the smaller turtle is not getting enough of the UV light provided by the basking area. This is a concern for the turtles health because there cold blood and they rely on the basking lamp to regulate their body heat. The UV light will also provide the Nutrient's the turtles need to keep their shells protected and healthy

The Primary stakeholders include Dr. Faith Weeks, Robert, and the rest of our living lab team. The secondary stakeholders are the students and other faculty members of Towson University

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Element F

Leland Greco

Design Viability & Risk Assessment:

Areas where we could have possible risks are:

- Building size
- Cost
- Construction time
- Heating lamps being too hot or too cold
- Electrical problems with wiring and turtle safety

Risk Mitigation Plan:

To address the possible risks, follow the same order

- Building size: We will take careful and proper measurements to ensure that it is properly integrated into the tank. While we want to build a new basking area, we will consider components we can make from cheaper materials, without jeopardizing the turtle's safety from possible ingestion

- Construction time: We will minimize the amount of time it takes for the construction of the new basking area, ensuring that we use cheaper and quicker materials to complete it accurately. Before moving on to a final iteration that will be the most time-consuming and cost-consuming

- Electrical problems:

Leland Greco

Due to possible ingestion issues, we will take careful and precise measurements to ensure the basking platform is properly integrated into the tank and does not obstruct anything. While we aim to build a new basking area, we will consider using certain components made from cheaper materials, without jeopardizing the turtles' safety.

Construction Time: We will minimize the construction time for the new basking area by selecting materials that are both cost-effective and quick to work with. This approach will allow us to efficiently reach a final design, while keeping in mind that the most cost-effective solution may also be more time-consuming to execute.

Electrical Problems: We need to address electrical concerns as part of our project. Having heat lamp right above water that needs to be plugged in there is always a danger of the electricity and the water making contact. To prevent this from becoming a problem we will added a mesh lid in top of our basking platform that the UV lights will be able to sit on top of, while being able to give the turtles heat and they UV rays they need and be in a safe area not making contact with the water

Our team is tasked with creating a second basking platform to support the smaller turtle, Zark. Our design must accommodate the weight of the turtles, provide ample space for them to fit comfortably, and ensure that the proper lighting and heat are available for their shell health and body temperature. During discussions with the primary stakeholder, Robert, he emphasized that our platform should not be made from any soft plastics that the turtles could bite or scratch through. Additionally, we need to design a ramp that turtles can grip easily and climb up. Therefore, we will be using a hard plastic with a textured surface for both the ramp and the flooring of our platform. We have credible evidence that our design will effectively function for our turtles. The Living Lab in Towson, where our primary stakeholder is based, supports our design approach.

Element H

This is our original prototype; we made it out of cardboard, hot glue, and a small amount of scotch tape. We chose to use cardboard because it was easily accessible and we had left over from another project. It also helped us keep the lightweight idea that our final design will keep.

Testing our design, there are multiple things we will be testing it against. Our main requirement is that we cannot harm the turtles in any way. Expanding the basking area so we are able to fit both of the turtles is a very high priority. Our testing should ensure that the area we create will be able to receive the same amount of heat in all areas of our basking area. Another one of our high-priority design requirements is making sure our design is able to be integrated into the system without disrupting the water temperature.

We will be able to test our product by placing our final 3D printed design in the tank, adding the correct lights on top, and placing a camera in the room that will be watching the turtles and see how they interact and use their new basking platform. During our testing, we will be watching to see how they interact and if they are able to associate their new platform with their old platform. We will be checking if they are using it correctly and are getting the light rays they need. We will also make sure that they both are able to fit and have enough space to move freely. Our goal is that they spend adequate time in their new platform, that their health increases from the more light they get, and that they are not fighting over space and are able to share their new platform.

Our testing plan will be able to test for every single priority that we have ranging from our highest priority to our lowest priority. We will test to make sure our product does not harm the turtles in any way or make their quality of living worse, We will make sure our testing proves that we expanded the basking are, by making sure both turtles can comfortable fit. We will be checking to see if each turtle is receiving the amount of heat and UV rays they need to maintain good health. We will also be checking the temperature of the water and making sure of design is able to be integrated without messing with the living of the turtles. Some of the lowest priority's that we will be testing for include, making sure our design will be able to fit on top of the tank and will not move around or fall off. We think this is a low priority because we know the measurements and have built our design accordingly.

We will run test to make sure the turtles are getting there proper UV rays and heat by checking them by their duration they stay in there basking platform and by using a thermometer to get the temperatures of their shell to make sure there body is at correct temperatures. We will also watch how the turtles interact with each other and see if they are still competing over the platform or if they both are able to sit in there new platform without fighting. To make sure our build is able to support the weight of both of the turtles we will place the basking area in the tank and add the combine body weight of both turtles in the platform will different objects to make sure it supports there weight. We will also be checking the temperature of the turtles water through the day to make sure our design does not interfere with the habitat.

We have had two different field experts come and check our design, Robert Towson lead of the living lab, said he liked our overall ideas about creation. He said that instead of making a smaller basking area just for Zark that instead we should just increase the size of the original platform so that both turtles would be able to fit comfortably. Mr. Jorge Gomez-Jurad our second field expert also liked our idea but had a few modifications that we added to our design. Some of the ideas he had that we incorporated were, added a non-slip surface to our ramp so the turtles were able to climb up without problems, adding a hinge and floatation device to the ramp so when the tank had to be emptied for cleaning or other maintenance it would make it easier and more efficient. He also had the same idea as Robert that we should make one larger basking platform instead keeping the one they have right and adding a smaller one.

Element L

During our design process, we have had many ideas and iterations in our design. Many of their ideas we would keep, and we would implement in our next design the texture on the floor and ramp, which both stakeholders said the turtles need for this platform to be useful and improve their lives. Another idea we will keep is making a single large platform instead of an additional platform just for the smaller turtles, which both stakeholders told my team was a better idea.

Another idea I would keep was the walls going alongside the ramps to provide the turtle with safety when getting into his basking area.

Some ideas I would have changed if I were to do this project again would be trying to add a divider so the turtles would not have to interact with each other and would be able to have their own space to eliminate the competition for space. Another item I would change would be the area of the enclosure. I would increase the size so they are able to move more freely and are less limited in space.

If I were to give advice for other students who are asked to approached a challenge similar to the questions we were asked I would recommend making many different prototypes and starting with ones that can be made quickly, so they are able to make changes and differ there design quickly and not wait to the end and make last minute changes. I would also recommend meeting with their primary stake holders often and ask them many questions on their view of the design and how they would change and improve it.

How I would implement their chances would be in increase the square footage of our design while keeping the light weight and the structural support. Also to add the divider I would incorporate it into the 3D Printer so there would be a barrier